

Supplementary Winter Examination – 2024

Course: B. Tech. Branch: Petrochemical Engineering Semester: VI

Subject Code & Name: Plant Utility and Process Safety (BTCHC605_Y21)

Max Marks: 60

Date: 30/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(CO)	Mark
Q. 1	Objective type questions. (Compulsory Question)		12
1	Water that form scum with soap is called..... Water. a. Hard b. Soft c. Distilled d. None	CO 1	1
2	Which of the following substances are commonly used in a filter? a. Charcoal b. Sand c. Both a & b d. None.	CO 1	1
3	The type of cooling tower with maximum heat transfer from air to water is Draft. a. Natural b. Mechanical c. Both a & b d. None	CO 2	1
4	Toxicants enter organism through a. Ingestion b. Injection c. Absorption d. All	CO 3	1
5	are effective in preventing eye injury from chemical splashes, impact, dusty environments and welding. a. Googles b. Ear muff c. Coat d. Face guard	CO 6	1
6	All electrical equipment before use. a. Clean b. Inspect c. Label d. Organize	CO 5	1
7	The extinguishing method by which a pressurized water extinguisher works is..... a. cooling b. oxygen c. fuel removal d. breaking the chemical chain reaction.	CO 6	1
8	Which one is the water tube boiler? a. Cochran b. Lancashire c. Babcock d. None Wilcox	CO 2	1
9	A device used for pumping water into the boiler is called..... a. Economizer b. Feed pump c. Superheater d. Air preheater	CO 3	1

10	Industrial safety management is that branch of management which is concerned with _____ hazards from the industries. a. Controlling b. Reducing c. Eliminating d. All	CO 4	1
11	Which are different types of accidents? a. Minor b. Serious c. Fatal d. All	CO 1	1
12	Injury without showing external signs isinjury. a. Internal b. External c. Permanent d. Temporary	CO 1	1
Q. 2	Solve the following.		12
A)	Define plant utility and write down the importance of plant utility.	CO 1	6
B)	Explain water storage and distribution for firefighting. Explain pollution abating by flare and vents.	CO 1	6
Q.3	Solve the following.		12
A)	Describe the construction and working of Simple vertical boiler in detail.	CO 2	6
B)	Describe in details Indian Boiler Act,1923. (Any 6 Points)	CO 2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain the vapor compression refrigeration cycle with neat diagram.	CO 3	6
B)	Write a note on a) Brine b) Antifreeze solution	CO 3	6
C)	Explain in detail fail safe concept.	CO 4	6
Q.5	Solve Any Two of the following.		12
A)	What are the safety parameters in process design? Explain polyvinyl chloride plant with its safety parameters.	CO 4	6
B)	Write down the hazard of metals: a) Arsenic b) Cadmium c) Mercury	CO 5	6
C)	Describe in details electrical safety in chemical process plant.	CO 5	6
Q. 6	Solve Any Two of the following.		12
A)	Describe with neat sketch any one type of fire extinguisher in detail.	CO 6	6
B)	What is safety checklist? Explain safety checklist for installation.	CO 6	6
C)	What are the protective devices used in industry for protection of a) Face b) Eyes c) Ears d) Legs e) Body f) Hand	CO 6	6

*** End ***